

1. (a) Write down the exact value of $\cos 405^\circ$

(1)

- (b) Hence, using a double angle identity for cosine, or otherwise, determine the exact value of $\cos 101.25^\circ$, giving your answer in the form

$$a\sqrt{b + c\sqrt{2 + \sqrt{2}}}$$

where a , b and c are rational numbers.

(5)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 6 marks)



2. A student is attempting to prove that there are infinitely many prime numbers.

The student's attempt to prove this is in the box below.

Assume there are only finitely many prime numbers, then there is a biggest prime number, p .

Let $n = 2p + 1$. Then n is bigger than p and since $2p + 1$ is not divisible by p , n is a prime number.

Hence n is a prime number bigger than p , contradicting the initial assumption. So we conclude there are infinitely many prime numbers.

- (a) Use $p = 7$ to show that the following claim made in the student's proof is **not** true:

since $2p + 1$ is not divisible by p , n is a prime number.

(1)

The student changes their proof to use $n = 6p + 1$ instead of $n = 2p + 1$

- (b) Show, by counter example, that this does not correct the student's proof.

(2)

- (c) Write out a correct proof by contradiction to show that there are infinitely many prime numbers.

(5)

(+S1)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 9 marks)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

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(Total for Question 3 is 10 marks)



4. (a) Use the trapezium rule with 4 strips to find an approximate value for

$$\int_0^1 16^x \, dx \quad (2)$$

- (b) Use the trapezium rule with n strips to write down an expression that would give an approximate value for

$$\int_0^1 16^x \, dx \quad (2)$$

- (c) Hence show that

$$\int_0^1 16^x \, dx = \lim_{n \rightarrow \infty} \left(\frac{1}{n} \left(1 + 16^{\frac{1}{n}} + \dots + 16^{\frac{n-1}{n}} \right) \right) \quad (3)$$

- (d) Use integration to determine the exact value of

$$\int_0^1 16^x \, dx \quad (3)$$

Given that the limit exists,

- (e) use part (c) and the answer to part (d) to determine the exact value of

$$\lim_{x \rightarrow 0} \frac{16^x - 1}{x} \quad (5)$$

(+S1)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 16 marks)



5.

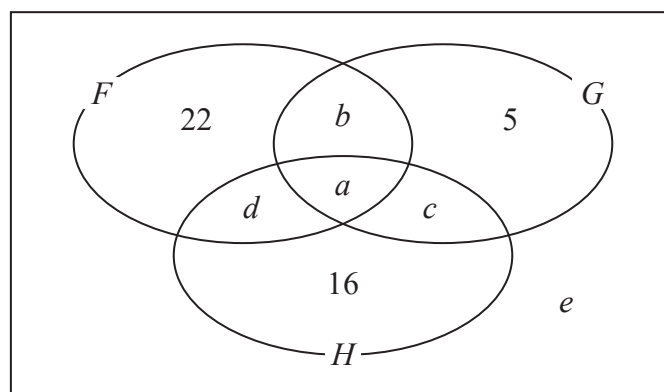


Figure 2

Figure 2 shows a partially completed Venn diagram of sports that a year group of students enjoy, where a , b , c , d and e are non-negative integers.

The diagram shows how many students enjoy a combination of football (F), golf (G) and hockey (H) or none of these sports.

There are n students in the year group.

It is known that

- $P(F) = \frac{3}{7}$
- $P(H | G) = \frac{1}{3}$
- F is independent of $H \cap G$

(a) Show that $P(F \cap H \cap G) = \frac{1}{7} P(G)$ (3)

(b) Prove that if two events X and Y are independent, then X' and Y are also independent. (3)

(c) Hence find the value k such that $P(F' \cap H \cap G) = kP(G)$ (2)

(d) Show that $c = \frac{4}{3}a$ (2)

Given further that $P(F | H) = \frac{1}{5}$

(e) find an expression for d in terms of a , and hence deduce the maximum possible value of a . (3)

(f) Determine the possible values of n . (6)
(+S2)



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Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing answers.

(Total for Question 5 is 21 marks)



6.

In this question you may assume the following formulae for the volume and curved surface area of a cone of base radius r and height h and of a sphere of radius r .

Cone: volume $V = \frac{1}{3}\pi r^2 h$ and curved surface area $S = \pi r \sqrt{h^2 + r^2}$

Sphere: volume $V = \frac{4\pi}{3}r^3$ and curved surface area $S = 4\pi r^2$

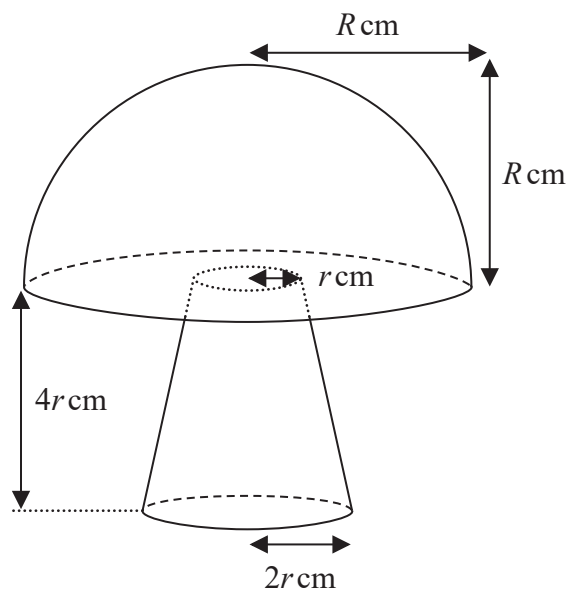


Figure 3

Figure 3 shows the design for a garden ornament.

The ornament is made of a hemisphere on top of a truncated cone.

The truncated cone has base radius $2r$ cm, top radius r cm and height $4r$ cm.

The hemisphere has radius R cm.

Given that the volume of the ornament is $2100\pi \text{ cm}^3$

(a) show that

$$R^3 = 3150 - 14r^3 \quad (5)$$

(b) Find an expression involving $\frac{dR}{dr}$ in terms of r and/or R .

(2)

The base of the truncated cone of the ornament is fixed to the ground.

(c) Show that the visible surface area of the ornament, $A \text{ cm}^2$, is given by

$$A = (3\sqrt{17} - 1)\pi r^2 + 3\pi R^2 \quad (5)$$



Question 6 continued

(d) Hence show that

$$\frac{dA}{dr} = \gamma\pi r - \frac{\delta\pi r^2}{R}$$

where γ and δ are real numbers to be determined.

(2)

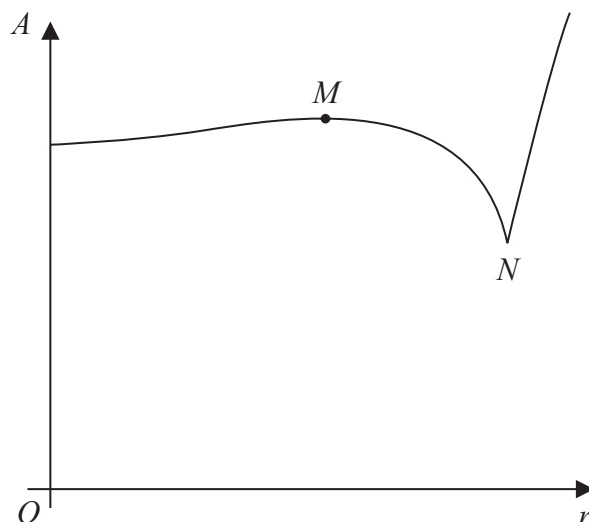


Figure 4

Figure 4 shows a sketch of A against r , for $r \geq 0$

There is a local minimum at $r = 0$ and a local maximum at the point M . The overall minimum point is at the point N , where the gradient of the curve is undefined.

(e) (i) Determine the r coordinate of the point N .

(ii) Explain why, for the ornament, r must be less than this value.

(2)

(f) Show that the r coordinate of the point M is

$$\sqrt[3]{\frac{p(3\sqrt{17}-1)^3}{3q^2 + (3\sqrt{17}-1)^3}}$$

where p and q are integers to be determined.

(5)
(+S2)

Question 6 continued

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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 23 marks)



7. A sequence of non-zero real numbers $a_1, a_2, a_3, \dots$ is defined by

$$a_{n+1} = p + \frac{q}{a_n} \quad n \in \mathbb{N}$$

where p and q are real numbers with $q \neq 0$

It is known that

- one of the terms of this sequence is a
- the sequence is periodic

(a) Determine an equation for q , in terms of p and a , such that the sequence is constant (of period/order one).

(2)

(b) Determine the value of p that is necessary for the sequence to be of period/order 2.

(3)

(c) Give an example of a sequence that satisfies the condition in part (b), but is **not** of period/order 2.

(2)

(d) Determine an equation for q , in terms of p only, such that the sequence has period/order 4.

(7)

(+S1)



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Question 7 continued

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Question 7 continued

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(Total for Question 7 is 15 marks)

TOTAL FOR PAPER IS 100 MARKS

